Circles and Parabolas

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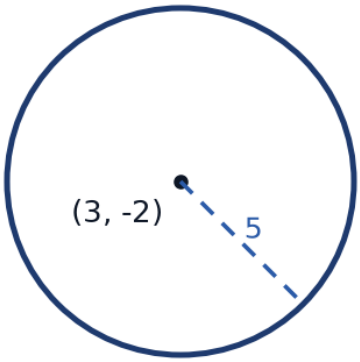
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Circles**

Write the standard-form equation of the circle with center $(3, -2)$ and radius 5.

2 **[Foundational]** • **Target: Circles**

Consider the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.



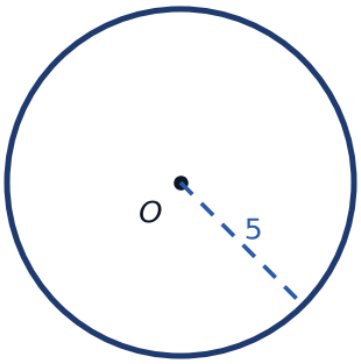
a Write the equation in standard form by completing the square.

b State the center and radius.

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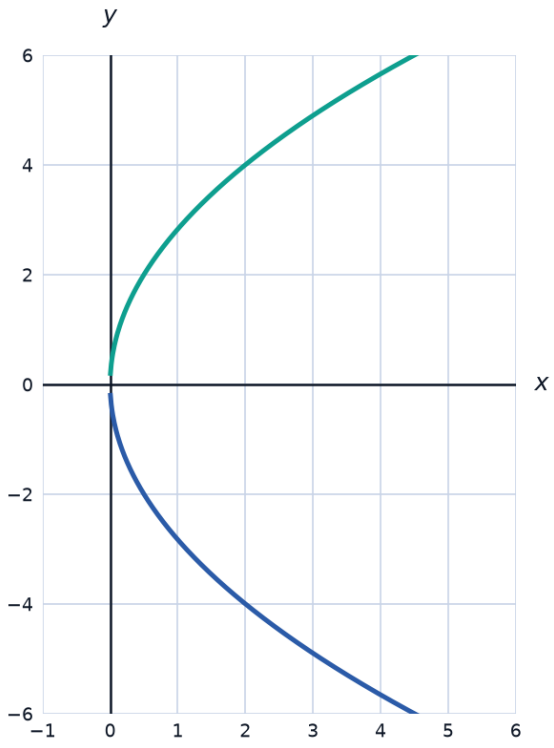
3 [Foundational] • Target: Circles

Determine whether the point $(6, 2)$ lies inside, on, or outside the circle $(x - 3)^2 + (y + 2)^2 = 25$.



4 [Foundational] • Target: Parabolas with vertex at the origin

For the parabola $y^2 = 8x$, find:



a the focus

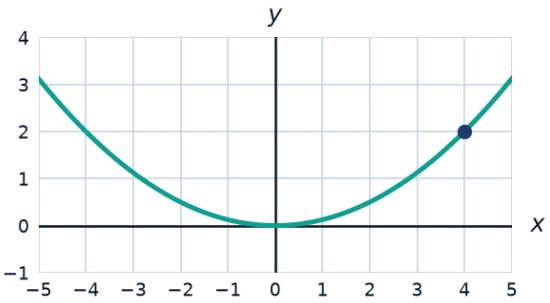
b the directrix

c the direction of opening

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5 [Foundational] • Target: Parabolas with vertex at the origin

A parabola has vertex at the origin, opens upward, and passes through the point $(4, 2)$.

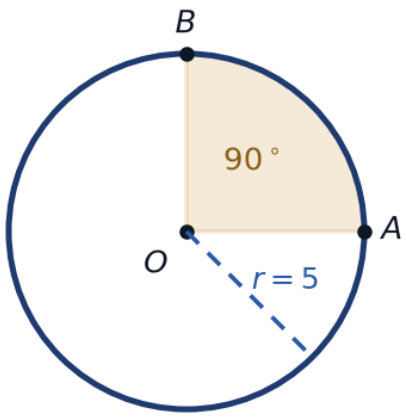


a Find its equation in the form $x^2 = 4py$.

b State the coordinates of the focus.

6 [Foundational] • Target: Parabolas with vertex at the origin

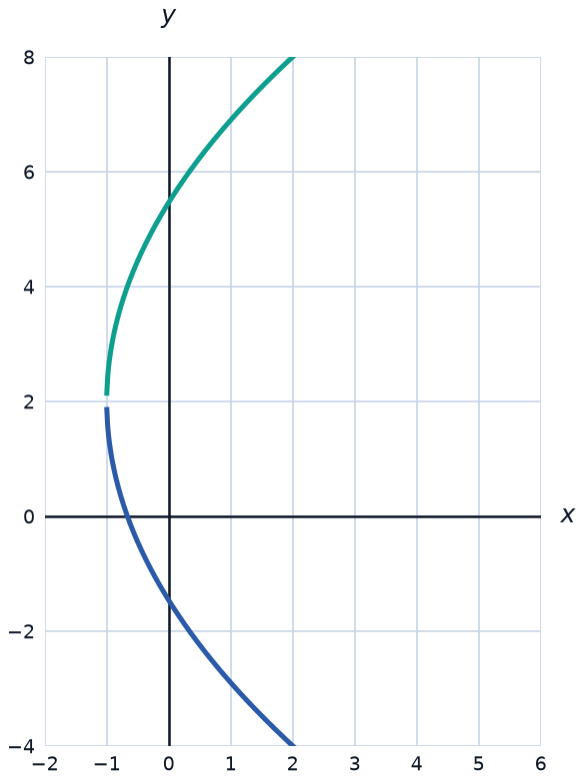
A satellite dish has a parabolic cross-section 2 m wide and 0.5 m deep. Taking the vertex at the origin opening upward, find the distance from the vertex to the focus (where the receiver is placed).



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7 [Foundational] • Target: Parabolas with vertex not at the origin

For the parabola $(y-2)^2 = 12(x+1)$, find:



a the vertex

b the value of p

c the focus

d the directrix

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Parabolas with vertex not at the origin

Write the equation of the parabola with vertex $(2, -3)$, opening downward, with $|p| = 1$.

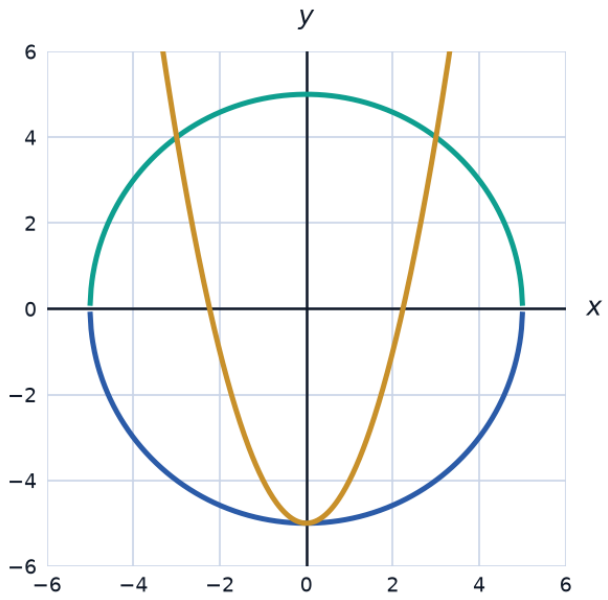
9 [Intermediate] • Target: Parabolas with vertex not at the origin

Complete the square to write $y = x^2 - 6x + 5$ in vertex form, and state the vertex.

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10 [Intermediate] • Target: Mixed practice: circles and parabolas together

A circle and a parabola are given by $x^2 + y^2 = 25$ and $y = x^2 - 5$. Find their points of intersection.



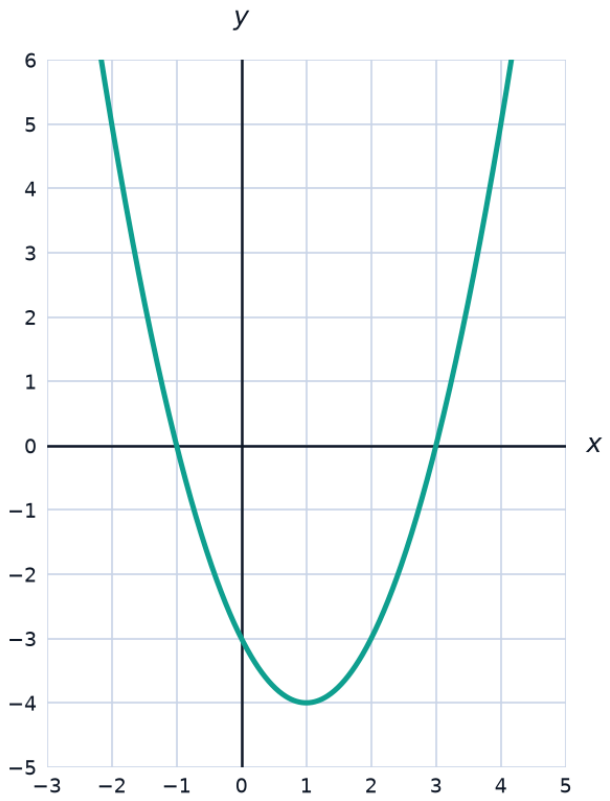
11 [Intermediate] • Target: Mixed practice: circles and parabolas together

A circular satellite orbit has the equation $x^2 + y^2 = (6400 + 500)^2$ (Earth radius plus altitude, in km). Find the radius of this orbit.

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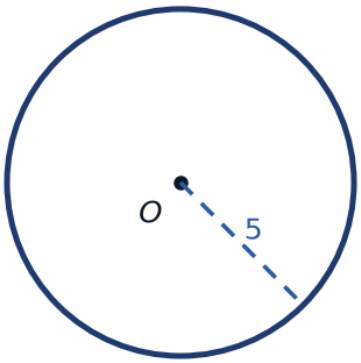
12 [Intermediate] • Target: Graphing circles and parabolas

The parabola $y = x^2 - 2x - 3$ is shown. Use the graph to estimate its vertex and x-intercepts, then verify algebraically.



13 [Intermediate] • Target: Tangent lines to circles

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.



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14 [Intermediate] • Target: The reflective property of parabolas

Explain the reflective property of a parabola that makes it useful for satellite dishes and headlights, in terms of the focus.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Finding a circle through three points • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Explain the general strategy for finding the equation of a circle passing through three given (non-collinear) points, without fully solving a specific example.

16 [Challenging] • Target: Distance from a point to the focus (parabola definition) • Cross-Lesson: 9.02 Geometric Sequences and Series

State the focus-directrix definition of a parabola, and use it to verify that the point $(4, 4)$ lies on $y^2 = 4x$ (focus $(1, 0)$, directrix $x = -1$).

17 [Challenging] • Target: Systems involving circles • Cross-Lesson: 9.03 The Binomial Theorem

Find the points of intersection of the circles $x^2 + y^2 = 25$ and $(x - 6)^2 + y^2 = 25$.

18 [Challenging] • Target: Mixed practice: circles and parabolas together • Cross-Lesson: 1.01 Functions and Function Notation

Consider the points $A(1, 2, 3)$, $B(2, 4, 1)$, and $C(-1, 0, 5)$ in $\mathbb{R}^3$. Find a non-zero vector $\mathbf{n}$ normal to the plane containing A, B, C, and write the Cartesian equation of the plane.

19 [Challenging] • Target: Graphing circles and parabolas • Cross-Lesson: 1.02 Domain and Range

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

20 [Challenging] • Target: Tangent lines to circles • Cross-Lesson: 1.03 Average Rate of Change

Write the standard equation of the ellipse having foci at $(\pm 3, 0)$ and vertices at $(\pm 5, 0)$. State the lengths of the major and minor axes, and compute the eccentricity.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Standard Equation and Features of Circles

Find the center and radius of the circle $(x-3)^2+(y+5)^2=49$.

a. Center: $(3,\ -5)$, Radius: 7

b. Center: $(-3, 5)$, Radius: 7

c. Center: $(3,\ -5)$, Radius: 49

d. Center: $(-3, 5)$, Radius: 49

22 [Foundational] • Target: Standard Equation and Features of Circles

Convert the general equation $x^2+y^2-6x+8y-11=0$ to standard form by completing the square.

a. $(x-3)^2+(y+4)^2=36$

b. $(x+3)^2+(y-4)^2=36$

c. $(x-3)^2+(y+4)^2=25$

d. $(x-6)^2+(y+8)^2=11$

23 [Foundational] • Target: Standard Equation and Features of Circles

Find the equation of the circle passing through the origin with center $(-2, 3)$.

a. $(x+2)^2+(y-3)^2=13$

b. $(x-2)^2+(y+3)^2=13$

c. $(x+2)^2+(y-3)^2=5$

d. $(x+2)^2+(y-3)^2=25$

24 [Foundational] • Target: Parabolas with Vertical Axis of Symmetry

Find the focus and directrix of the parabola $x^2=12y$.

a. Focus: $(0, 3)$, Directrix: $y=-3$

b. Focus: $(3, 0)$, Directrix: $x=-3$

c. Focus: $(0, 12)$, Directrix: $y=-12$

d. Focus: $(0, 6)$, Directrix: $y=-6$

25 [Foundational] • Target: Parabolas with Vertical Axis of Symmetry

Find the standard equation of the parabola with vertex $(0, 0)$ and focus $(0,\ -2)$.

a. $x^2=-8y$

b. $y^2=-8x$

c. $x^2=8y$

d. $y^2=-4x$

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26 [Foundational] • Target: Parabolas with Vertical Axis of Symmetry

Find the focal diameter (latus rectum length) of $(x-1)^2 = -16(y+2)$.

a. 16

b. 4

c. 8

d. 32

27 [Foundational] • Target: Parabolas with Horizontal Axis of Symmetry

Find the vertex, focus, and directrix of $(y-2)^2 = 8(x+1)$.

a. Vertex: $(-1, 2)$, Focus: $(1, 2)$, Directrix: $x = -3$

b. Vertex: $(1, -2)$, Focus: $(3, -2)$, Directrix: $x = -1$

c. Vertex: $(-1, 2)$, Focus: $(-1, 4)$, Directrix: $y = 0$

d. Vertex: $(2, -1)$, Focus: $(4, -1)$, Directrix: $x = 0$

28 [Foundational] • Target: Parabolas with Horizontal Axis of Symmetry

Find the equation of a parabola with vertex $(3, -1)$ and focus $(1, -1)$.

a. $(y+1)^2 = -8(x-3)$

b. $(x-3)^2 = -8(y+1)$

c. $(y+1)^2 = 8(x-3)$

d. $(y-3)^2 = -4(x+1)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Parabolas with Horizontal Axis of Symmetry

Convert $y^2 + 4y - 4x + 8 = 0$ to standard form.

a. $(y+2)^2 = 4(x-1)$

b. $(y-2)^2 = 4(x+1)$

c. $(y+2)^2 = 4x - 8$

d. $(x-1)^2 = 4(y+2)$

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30 [Intermediate] • Target: Reflective Properties and Applications

A satellite dish has the shape of a paraboloid. If the cross-section is $x^2 = 20y$ (in inches), where should the receiver be placed?

a. At $(0, 5)$, which is 5 inches above the vertex

b. 20 inches above the vertex

c. 10 inches above the vertex

d. 2.5 inches above the vertex

31 [Intermediate] • Target: Reflective Properties and Applications

A suspension bridge cable hangs in the shape of a parabola with towers 100 m apart and 20 m high, touching the road at the center. Find its equation with origin at the vertex.

a. $y = \frac{1}{125}x^2$

b. $y = \frac{1}{250}x^2$

c. $y = \frac{1}{50}x^2$

d. $y = \frac{1}{20}x^2$

32 [Intermediate] • Target: Reflective Properties and Applications

Why do parabolic headlights use a light source placed exactly at the focus?

a. Light rays reflecting off the parabolic surface emerge in a parallel beam

b. It spreads the light in all directions equally

c. It minimizes heat production

d. It prevents the mirror from tarnishing

33 [Intermediate] • Target: Parabolas with Horizontal Axis of Symmetry

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y-coordinate of the point P?

a. 3

b. 4

c. 5

d. 6

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34 [Intermediate] • Target: Reflective Properties and Applications

A circle has its center at the origin $(0, 0)$ and passes through the point $P(3, y)$. If the equation of the circle is $x^2 + y^2 = 25$, what is the positive value of y?

a. 3

b. 4

c. 5

d. 16

35 [Intermediate] • Target: Circles

A circle is centered at the origin $(0, 0)$ and passes through the point $(3, 4)$. What is the equation of this circle?

a. $x^2 + y^2 = 7$

b. $x^2 + y^2 = 9$

c. $x^2 + y^2 = 16$

d. $x^2 + y^2 = 25$

36 [Intermediate] • Target: Parabolas with vertex at the origin

Consider a circle with its center at the origin $(0, 0)$ and a radius of 5 units. A chord of this circle is perpendicular to the x-axis and intersects the x-axis at the point $(3, 0)$. The endpoints of this chord on the circle are A and B. Determine the length of the chord AB.

a. 6

b. 8

c. 10

d. 4

37 [Intermediate] • Target: Parabolas with vertex not at the origin

A ramp rises 3 m over a horizontal distance of 20 m. Find its angle of inclination, to the nearest tenth of a degree.

a. 8.5°

b. 81.5°

c. 13.5°

d. 17.1°

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Mixed practice: circles and parabolas together • Cross-Lesson: 1.01 Functions and Function Notation

For the parabola $y^2 = -20x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(-5, 0)$, directrix $x = 5$

b. Vertex $(0, 0)$, focus $(-10, 0)$, directrix $x = 10$

c. Vertex $(0, 0)$, focus $(0, -5)$, directrix $y = 5$

d. Vertex $(-5, 0)$, focus $(0, 0)$, directrix $x = -5$

39 [Challenging] • Target: Graphing circles and parabolas • Cross-Lesson: 1.02 Domain and Range

For the parabola $y^2 = -16x$, find the vertex, focus, and directrix.

a. Vertex $(-4, 0)$, focus $(0, 0)$, directrix $x = -4$

b. Vertex $(0, 0)$, focus $(-8, 0)$, directrix $x = 8$

c. Vertex $(0, 0)$, focus $(0, -4)$, directrix $y = 4$

d. Vertex $(0, 0)$, focus $(-4, 0)$, directrix $x = 4$

40 [Challenging] • Target: Tangent lines to circles • Cross-Lesson: 1.03 Average Rate of Change

For the parabola $y^2 = -12x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(-6, 0)$, directrix $x = 6$

b. Vertex $(0, 0)$, focus $(0, -3)$, directrix $y = 3$

c. Vertex $(0, 0)$, focus $(-3, 0)$, directrix $x = 3$

d. Vertex $(-3, 0)$, focus $(0, 0)$, directrix $x = -3$

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41 [Challenging] • Target: The reflective property of parabolas • Cross-Lesson: 2.01 Right-Triangle Trigonometry

For the parabola $y^2 = -8x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(-4, 0)$, directrix $x = 4$

b. Vertex $(0, 0)$, focus $(0, -2)$, directrix $y = 2$

c. Vertex $(0, 0)$, focus $(-2, 0)$, directrix $x = 2$

d. Vertex $(-2, 0)$, focus $(0, 0)$, directrix $x = -2$

42 [Challenging] • Target: Finding a circle through three points • Cross-Lesson: 2.02 Radian Measure and Arc Length

For the parabola $y^2 = -4x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(-2, 0)$, directrix $x = 2$

b. Vertex $(-1, 0)$, focus $(0, 0)$, directrix $x = -1$

c. Vertex $(0, 0)$, focus $(0, -1)$, directrix $y = 1$

d. Vertex $(0, 0)$, focus $(-1, 0)$, directrix $x = 1$

43 [Challenging] • Target: Distance from a point to the focus (parabola definition) • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

For the parabola $y^2 = 20x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(0, 5)$, directrix $y = -5$

b. Vertex $(5, 0)$, focus $(0, 0)$, directrix $x = 5$

c. Vertex $(0, 0)$, focus $(10, 0)$, directrix $x = -10$

d. Vertex $(0, 0)$, focus $(5, 0)$, directrix $x = -5$

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44 [Challenging] • Target: Systems involving circles • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

For the parabola $y^2 = 12x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(3, 0)$, directrix $x = -3$

b. Vertex $(3, 0)$, focus $(0, 0)$, directrix $x = 3$

c. Vertex $(0, 0)$, focus $(6, 0)$, directrix $x = -6$

d. Vertex $(0, 0)$, focus $(0, 3)$, directrix $y = -3$

45 [Challenging] • Target: Standard Equation and Features of Circles • Cross-Lesson: 3.01 The Unit Circle and Exact Values

For the parabola $y^2 = 16x$, find the vertex, focus, and directrix.

a. Vertex $(0, 0)$, focus $(4, 0)$, directrix $x = -4$

b. Vertex $(4, 0)$, focus $(0, 0)$, directrix $x = 4$

c. Vertex $(0, 0)$, focus $(8, 0)$, directrix $x = -8$

d. Vertex $(0, 0)$, focus $(0, 4)$, directrix $y = -4$

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